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Okumura

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(54) **NAVIGATION SYSTEM**

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(51) **Int. Cl.**

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B60K 35/28	(2024.01)
B60W 30/095	(2012.01)
B60W 40/04	(2006.01)
B60W 50/14	(2020.01)

(52) **U.S. Cl.**

CPC ***B60W 50/14*** (2013.01); ***B60K 35/28*** (2024.01); ***B60W 30/0956*** (2013.01); ***B60W 40/04*** (2013.01); *B60K 2360/178* (2024.01); *B60W 2050/146* (2013.01); *B60W 2554/406* (2020.02); *B60W 2555/20* (2020.02); *B60W 2556/10* (2020.02); *B60W 2556/45* (2020.02)

(58) **Field of Classification Search**

CPC .. B60W 50/14; B60W 30/0956; B60W 40/04; B60W 2050/146; B60W 2554/406; B60W 2555/20; B60K 35/28; B60K 2360/178

USPC 340/438

See application file for complete search history.

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(57) **ABSTRACT**

A navigation system, including: a memory; a display; and a processor coupled to the memory and to the display, the processor being configured to: at a time of route guidance for a host vehicle, estimate an aspect of an accident at a vicinity of the route and a risk of the accident; and control the display so as to display objects indicating a collision target corresponding to the aspect of the accident, with a number of the objects corresponding to a level of the risk.

5 Claims, 5 Drawing Sheets

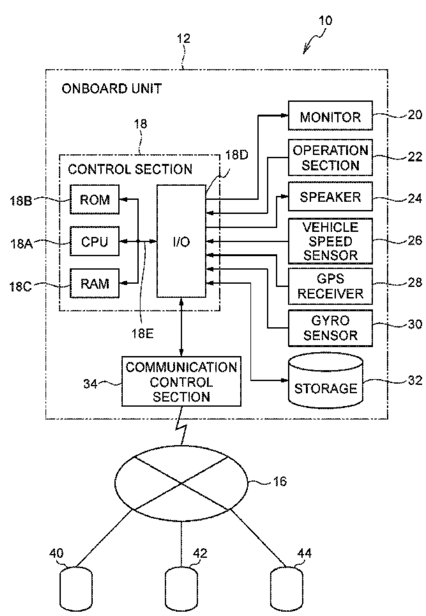

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FIG.1

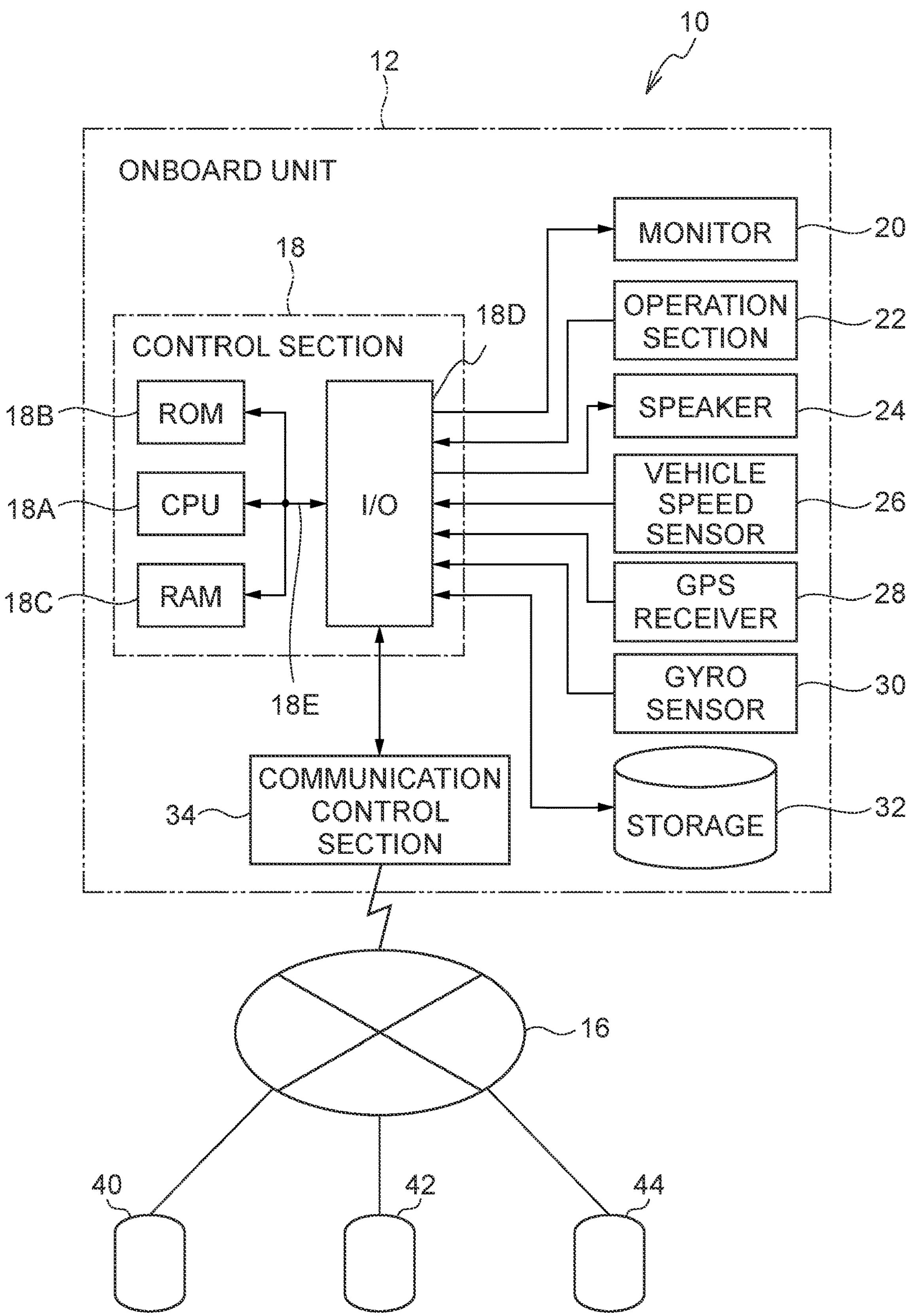

10
12
ONBOARD UNIT
18
CONTROL SECTION
18D
18B
ROM
18A
CPU
18C
RAM
I/O
18E
MONITOR
20
OPERATION SECTION
22
SPEAKER
24
VEHICLE SPEED SENSOR
26
GPS RECEIVER
28
GYRO SENSOR
30
STORAGE
32
COMMUNICATION CONTROL SECTION
34
16
40
42
44

FIG.2

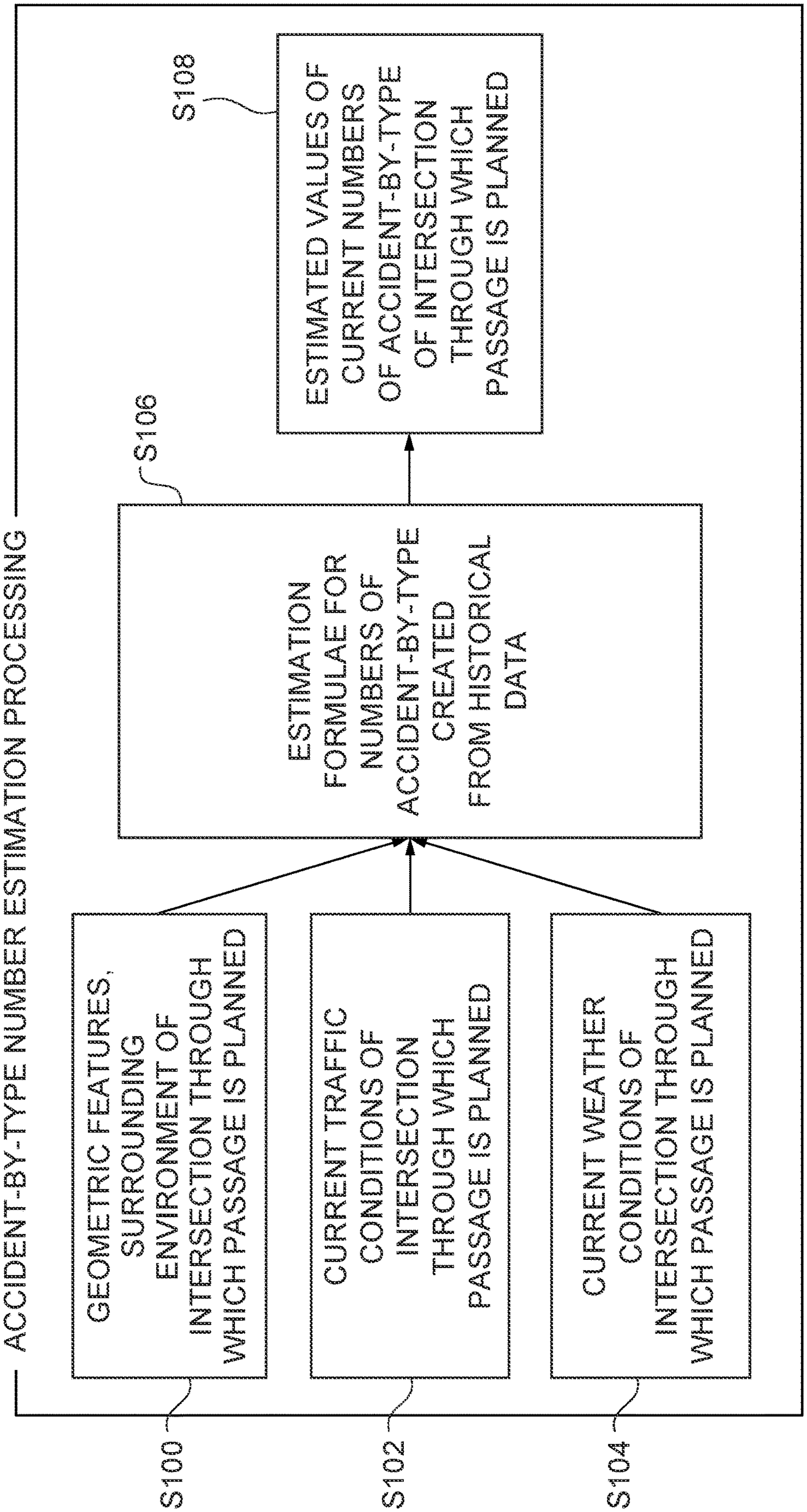

ACCIDENT-BY-TYPE NUMBER ESTIMATION PROCESSING
S100
GEOMETRIC FEATURES, SURROUNDING ENVIRONMENT OF INTERSECTION THROUGH WHICH PASSAGE IS PLANNED
S102
CURRENT TRAFFIC CONDITIONS OF INTERSECTION THROUGH WHICH PASSAGE IS PLANNED
S104
CURRENT WEATHER CONDITIONS OF INTERSECTION THROUGH WHICH PASSAGE IS PLANNED
S106
ESTIMATION FORMULAE FOR NUMBERS OF ACCIDENT-BY-TYPE CREATED FROM HISTORICAL DATA
S108
ESTIMATED VALUES OF CURRENT NUMBERS OF ACCIDENT-BY-TYPE OF INTERSECTION THROUGH WHICH PASSAGE IS PLANNED

FIG.3A

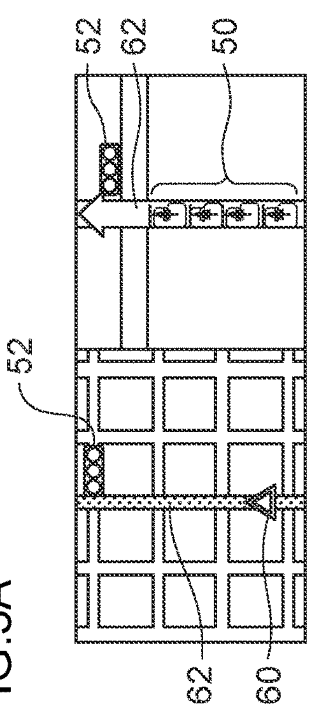


FIG.3B

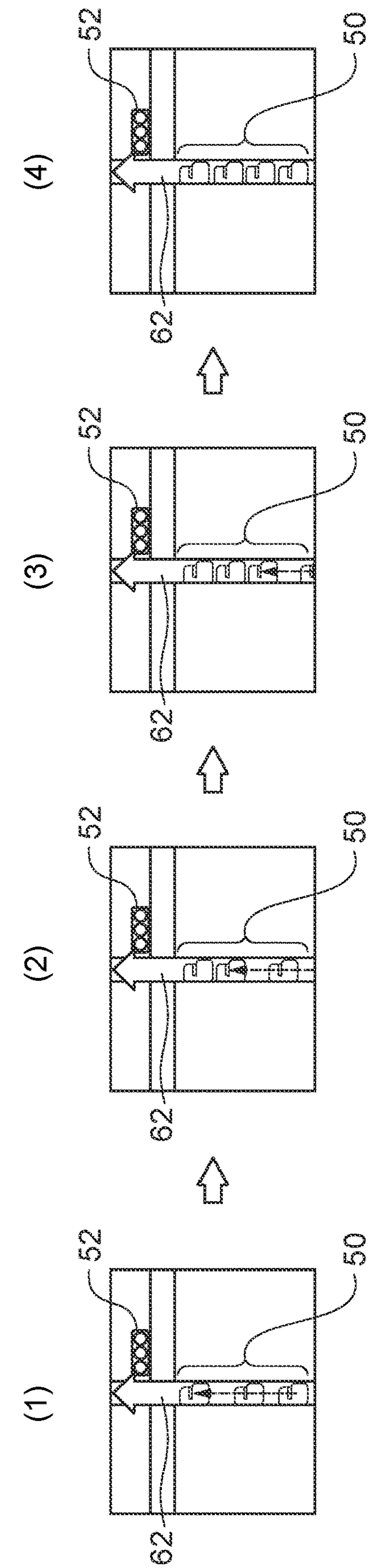


FIG.4A

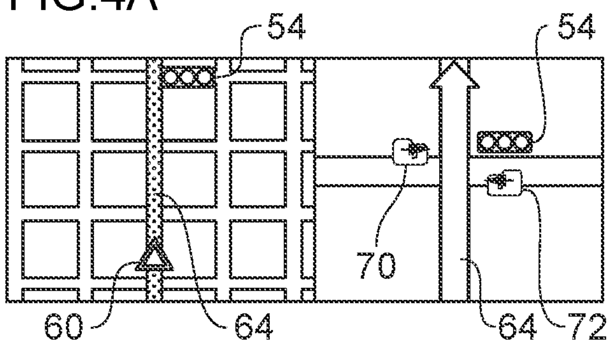


FIG.4B

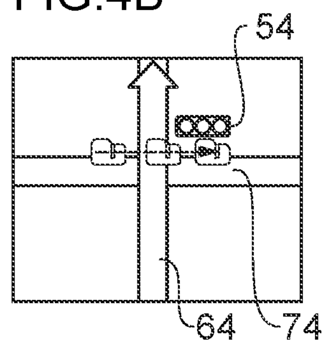


FIG.4C

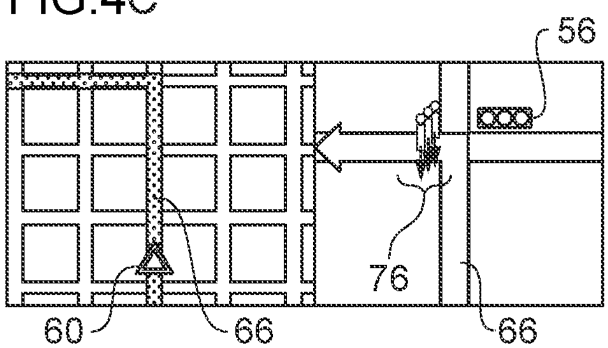


FIG.4D

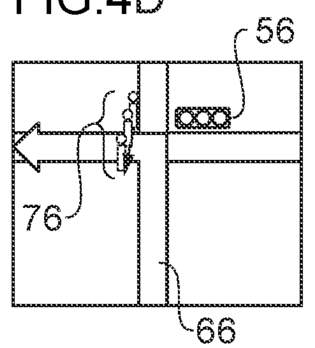


FIG.4E

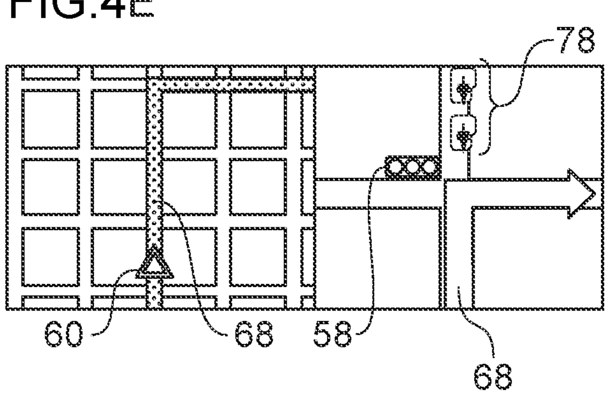


FIG.4F

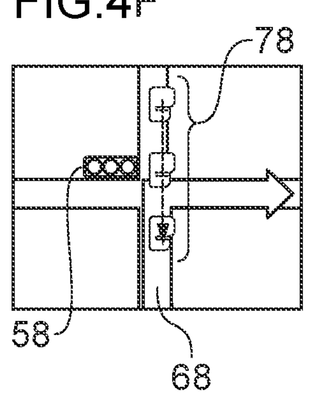


FIG.5A

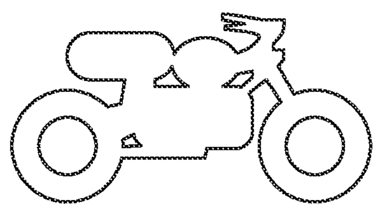
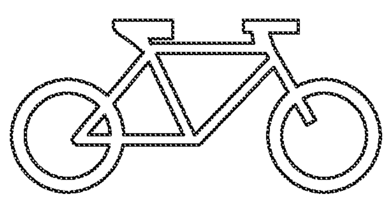


FIG.5B



NAVIGATION SYSTEM

CROSS-REFERENCE TO RELATED APPLICATION

This application claims priority under 35 USC 119 from Japanese Patent Application No. 2022-206028, filed on Dec. 22, 2022, the disclosure of which is incorporated by reference herein.

BACKGROUND

Technical Field

The present disclosure relates to a navigation system capable of displaying a risk of accident.

Related Art

In route guidance using a navigation system, there are cases in which high-accident location exist along a route.

Japanese Patent Application Laid-open (JP-A) No. 2006-163973, discloses a safe driving assistance system that presents a driver with places requiring caution in terms of traffic safety.

However, although the system described in JP-A No. 2006-163973, uses text, symbols, or the like to indicate places requiring caution on travel route, the level of risk of such places requiring caution is hard to be visually recognized.

SUMMARY

The present disclosure provides a navigational system that enables visual recognition of a risk of an accident on a route.

A first aspect of the present disclosure is a navigation system that includes: a memory; a display; and a processor coupled to the memory and to the display, the processor being configured to: at a time of route guidance for a host vehicle, estimate an aspect of an accident at a vicinity of the route and a risk of the accident; and control the display so as to display objects indicating a collision target corresponding to the aspect of the accident, with a number of the objects corresponding to a level of the risk.

According to the navigation system of the first aspect of the present disclosure, by displaying objects indicating a collision target with a number corresponding to the estimated level of the risk, the risk of an accident on a route may be made visually recognized.

In a second aspect of the present disclosure, in the first aspect, the processor may be configured to display the objects with animation that changes position at a screen of the display, and increases the number of the objects in a case in which the risk of the accident is higher.

According to the navigation system of the second aspect of the present disclosure, by displaying objects indicating the collision target using an animation with a high visual effect, and displaying more objects the higher the risk of the accident is, the risk of an accident on the route may be made visually recognized.

In a third aspect of the present disclosure, in the second aspect, in accordance with the aspect of the accident, the animation may be displayed so as to be one or more of, head-on, crossing, or side-on, with respect to the host vehicle on the route.

According to the navigation system of the third aspect of the present disclosure, by animation corresponding to the type of accident, the risk of the accident on the route may be made visually recognized.

In a fourth aspect of the present disclosure, in the above aspects, the processor may be configured to estimate the aspect of the accident and the risk of the accident, by using at least one of geometric features of the route, information on a surrounding environment of the route, information on traffic conditions on the route, and weather information of the route, acquired from an external source.

According to the navigation system of the fourth aspect of the present disclosure, the current risk of the accident may be estimated based on the latest information acquired from the external source.

In a fifth aspect of the present disclosure, in the fourth aspect, the processor may be configured to estimate the aspect of the accident and the risk of the accident, by using a learned model obtained by machine learning using, as training data, an aspect of an accident and a risk of an accident corresponding to at least one of historical accident records, traffic conditions, or weather conditions.

According to the navigation system of the fifth aspect of the present disclosure, by appropriate machine learning, a system that may prompt an user to visually recognize the risk of the accident on the rout, may be constructed.

According to the above aspects, the navigational system according to the present disclosure may prompt the user to visually recognize the risk of the accident on the route.

BRIEF DESCRIPTION OF THE DRAWINGS

Exemplary embodiments will be described in detail based on the following figures, wherein:

FIG. **1** is a block diagram illustrating configuration of a navigation system according to the present exemplary embodiment;

FIG. **2** is a functional block diagram illustrating accident-by-type estimation processing performed by a control section of a navigation system according to the present exemplary embodiment;

FIG. **3**A is an example of a display of a monitor in a case in which a host vehicle travels straight ahead and passes through an intersection with a traffic signal;

FIG. **3**B is a diagram illustrating changes in a time series of the display of objects at the intersection with the traffic signal;

FIG. **4**A is an example of a display of a monitor in a case in which the host vehicle travels straight ahead and passes through the intersection with the traffic signal;

FIG. **4**B is an example of an animated display in a case in which the host vehicle travels straight ahead and passes through the intersection with the traffic signal;

FIG. **4**C is an example of a display of a monitor in a case in which the host vehicle turns left and passes through the intersection with the traffic signal;

FIG. **4**D is an example of an animated display in a case in which the host vehicle turns left and passes through the intersection with the traffic signal;

FIG. **4**E is an example of a display of a monitor in a case in which the host vehicle turns right and passes through the intersection with the traffic signal;

FIG. **4**F is an example of an animated display in a case in which the host vehicle turns right and passes through the intersection with the traffic signal;

FIG. **5**A is an example of an object indicating a motorcycle; and

FIG. **5**B is an example of an object indicating a bicycle.

DETAILED DESCRIPTION

Explanation follows regarding a navigation system **10** according to the present exemplary embodiment, with reference to FIG. **1**. The navigation system **10** illustrated in FIG. **1** is a server network system in which, an onboard unit **12** installed in a vehicle, a map database (DB) server **40**, a traffic information server **42**, and a weather information server **44**, are connected to a network **16**.

As illustrated in FIG. **1**, the onboard unit **12** includes a control section **18** configured by a computer including a central processing unit (CPU) **18**A, read only memory (ROM) **18**B, random access memory (RAM) **18**C, and an input/output interface (I/O) **18**D, connected to a bus **18**E.

The ROM **18**B stores a program or the like for route guidance to a destination. The program stored in the ROM **18**B is expanded in the RAM **18**C and executed by the CPU **18**A to perform control for route guidance to a destination.

The I/O **18**D is connected to a monitor **20**, an operation section **22**, a speaker **24**, a vehicle speed sensor **26**, a global positioning system (GPS) receiver **28**, a gyro sensor **30**, a storage **32**, and a communication control section **34**. Note that the I/O **18** D and the respective sections may be connected through an onboard network such as a controller area network (CAN).

The monitor **20** is provided on a center console or the like inside the vehicle cabin, and displays a map or the like for route guidance to a set destination.

The operation section **22** includes a touch panel, a switch, or the like for inputting setting of the destination, and the like. Operating the operation section **22** enables registration of a home point, setting of the destination, instructions to re-search the route, and the like.

The speaker **24** generates sounds such as speech or audio for route guidance to the set destination.

The vehicle speed sensor **26** detects a travel speed (hereafter referred to as vehicle speed) of the vehicle, and outputs the detection result to the control section **18**.

The GPS receiver **28** receives a GPS signal including time information from a GPS satellite, and outputs the received result to the control section **18**. The control section **18** calculates the vehicle position based on the plural GPS signals, and obtains vehicle position information.

The gyro sensor **30** detects acceleration, angular acceleration, angular velocity, and the like, and outputs the detection results to the control section **18**. This enables the control section **18** to detect the vehicle attitude based on the detection results of the gyro sensor **30**.

The storage **32** stores map information for performing route guidance to the set destination, and the control section **18** performs route guidance to the set destination based on the map information stored in the storage **32**. The storage **32** also stores a travel history, consisting of position calculation results of the host vehicle position, as a path history.

The communication control section **34** communicates with the network **16** to exchange information with each of the map DB server **40**, the traffic information server **42**, and the weather information server **44**.

FIG. **2** is a functional block diagram illustrating an example of accident-by-type estimation processing, performed by the control section **18** of the navigation system according to the present exemplary embodiment. At step S**100**, the geometric characteristics of an intersection through which passage is planned, and information relating to the peripheral environment around the intersection, are acquired from the map DB server **40**.

At step S**102**, information relating to current traffic conditions at the intersection through which passage is planned, is acquired from the traffic information server **42**.

At step S**104**, information relating to current weather conditions at the intersection through which passage is planned, is acquired from the weather information server **44**.

At step S**106**, estimation formulae to estimate the numbers of accident-by-type are constructed using learned models. The learned models are acquired by machine learning, using an aspect of an accident and a risk of accident, with respect to at least one of a record of historical accidents, traffic conditions or weather conditions, as training data.

At step S**108**, the estimation formulae for numbers of accident-by-type, and the information acquired from each of the map DB server **40**, the traffic information server **42**, and the weather information server **44**, are used to output estimated values of current numbers of accident-by-type, at the intersection through which passage is planned. The estimated values of the numbers of accident-by-type are used as an indicator of the level of risk of traffic accidents, as described below.

In the present exemplary embodiment, in a case in which the aspect of passage of the host vehicle at the intersection, matches with the high-risk type of traffic accident, an animation is displayed on the monitor **20**, in which number of objects simulating the collision target moves. Here, the number corresponds to the level of the risk.

FIG. **3**A is an example of a display on the monitor **20**, in a case in which the host vehicle **60** travels, as indicated by the travel path **62**, straight ahead and passes through the intersection with a traffic signal **52**. The left side of FIG. **3**A is the map display of route guidance, and the right side of FIG. **3**A is an animation illustrating an enlarged display of the vicinity of the intersection.

In a case in which the estimated values of the numbers of accident-by-type output at step S**108** indicate that the risk of a rear-end collision is high, at the intersection with the traffic signal **52**, in the present exemplary embodiment, as illustrated in the right side of FIG. **3**A, an object **50** is displayed as a collision target in which the host vehicle **60** might collide. Although plural objects **50** are displayed in FIG. **3**A, in the present exemplary embodiment, in a case in which the estimated numbers of accident-by-type is high, the number of displayed objects **50** becomes larger.

FIG. **3**B is an explanatory diagram illustrating examples of changes in a time series of displays of the objects **50** at the intersection with the traffic signal **52**. In the present exemplary embodiment, number of objects **50** that corresponds to the risk represented by the estimated numbers of accident-by-type is not displayed all at once, and rather, as illustrated in (**1**) of FIG. **3**B, first, a situation is displayed in which the first object **50** approaches the intersection, and next, as illustrated in (**2**) of FIG. **3**B, a situation in which another object **50** approaches the intersection is displayed. Moreover, as illustrated in (**3**) of FIG. **3**B, objects **50** are added in accordance with the risk, and finally, as illustrated in (**4**) of FIG. **3**B, four objects **50** are displayed.

As illustrated in FIG. **3**B, by displaying an animation of a situation in which the collision target object **50** approaches the intersection with one or more of the objects **50**, drivers may be alerted against rear-end collisions.

FIG. **4**A is an example of a display on the monitor **20**, in a case in which the host vehicle **60** travels straight ahead and passes through an intersection with a traffic signal **54**, as indicated by travel path **64**. In a case in which the estimated values of the numbers of accident-by-type indicate that there is a high risk of a side-on collision at the intersection, in the present exemplary embodiment, as illustrated in the right-hand side of FIG. **4**A, objects **70**, **72** simulating other vehicles crossing the travel path **64** are displayed in numbers corresponding to the risk.

FIG. **4**B is an example of an animation display, in a case in which the host vehicle **60** travels straight ahead and passes through the intersection with the traffic signal **54**, as indicated by the travel path **64**. Since the object **74** that simulates another vehicle crossing the travel path **64** is shown crossing the travel path **64** of the host vehicle **60**, drivers may be alerted against side-on collisions.

FIG. **4**C is an example of a display on the monitor **20**, in a case in which the host vehicle **60** turns left and passes through the intersection with a traffic signal **56**, as indicted by the travel path **66**. In a case in which the estimated values of the numbers of accident-by-type indicate that there is a high risk of a left-turn accident at the intersection, in the present exemplary embodiment, as illustrated in the right-hand side of FIG. **4**C, the object **76** simulating a pedestrian crossing the travel path **66** is displayed in a number corresponding to the risk.

FIG. **4**D is an example of an animation display, in a case in which the host vehicle **60** turns left and passes through an intersection with the traffic signal **56**, as indicated by the travel path **66**. Since the object **76** simulating a pedestrian crossing the travel path **66** is shown crossing the travel path **66** of the host vehicle **60** at the intersection with the traffic signal **56**, drivers may be alerted against left-turn accidents.

FIG. **4**E is an example of a display on the monitor **20**, in a case in which the host vehicle **60** turns right and passes through the intersection with a traffic signal **58**, as indicated by the travel path **68**. In a case in which the estimated values of the numbers of accident-by-type indicate that there is a high risk of a right-turn accident at the intersection having the traffic signal **58**, in the present exemplary embodiment, as illustrated in the right-hand side of FIG. **4**E, the object **78** simulating another vehicle traveling in the opposite direction in the travel path **68** is displayed in a number corresponding to the risk.

FIG. **4**F is an example of an animation display, in a case in which the host vehicle **60** turns right and passes through the intersection with the traffic signal **58**, as indicated by the travel path **68**. Since an object **78** simulating another vehicle traveling in the opposite direction in the travel path **68** is illustrated as entering the intersection with the traffic signal **58**, drivers may be alerted against right-turn accidents.

In the present exemplary embodiment, in addition to the objects **50**, **70** and the like simulating another vehicle, and the object **76** simulating a crossing pedestrian, respective objects simulating a motorcycle and a bicycle may be displayed in an animation.

FIG. **5**A is an example of the object simulating the motorcycle, and FIG. **5**B is an example of the object simulating the bicycle. In a case in which the estimated values of the numbers of accident-by-type output at step S**108** in FIG. **2** indicate that there is a possibility of a collision with a motorcycle or a bicycle, animation display of the object illustrated in FIG. **5**A or FIG. **5**B enables the driver to be alerted.

As described above, according to the navigation system **10** of the present exemplary embodiment, by displaying objects that indicate the collision target according to the type of accident, such as head-on, crossing, or side-on, with respect to the host vehicle on a route, in a number corresponding to the estimated level of risk, the risk of the accident on the route may be made visually recognized.

Moreover, by displaying objects that indicate the collision target using an animation with a high visual effect, and also displaying more objects as the risk of the accident increases, the risk of an accident on a route may be quickly and visually recognized.

Moreover, based on the latest information acquired from external servers, such as the map DB server **40**, the traffic information server **42**, and the weather information server **44**, the risk of the accident is estimated, and by displaying animation of objects according to the type of accident, the risk of the accident at the present time may be visually recognized.

What is claimed is:

1. A navigation system, comprising:
a memory;
a display; and
a processor coupled to the memory and to the display, the processor being configured to:
at a time of route guidance for a host vehicle, estimate an aspect of an accident at a vicinity of the route and a risk of the accident; and
control the display so as to display objects indicating a collision target corresponding to the aspect of the accident, with a number of the objects corresponding to a level of the risk;
estimate the aspect of the accident and the risk of the accident, by using at least one of geometric features of the route, information on a surrounding environment of the route, information on traffic conditions on the route, and weather information of the route, acquired from an external source; and
estimate the aspect of the accident and the risk of the accident, by using a learned model obtained by machine learning using, as training data, the aspect of the accident and the risk of the accident corresponding to at least one of historical accident records, traffic conditions, or weather conditions.

2. The navigation system of claim **1**, wherein the processor is configured to display the objects with animation that changes position at a screen of the display, and increases the number of the objects in a case in which the risk of the accident is higher.

3. The navigation system of claim **2**, wherein, in accordance with the aspect of the accident, the animation is displayed so as to be one or more of, head-on, crossing, or side-on, with respect to the host vehicle on the route.

4. The navigation system of claim **1**, wherein the processor is configured to display the objects at a different position of the display, according to the aspect of the accident.

5. The navigation system of claim **1**, wherein the processor is configured to:
in a case in which the aspect of the accident corresponds to a rear-end collision accident at an intersection, display the objects simulating another vehicle at a position at a near side of the intersection;
in a case in which the aspect of the accident corresponds to a side-on collision at the intersection, display the objects simulating another vehicle at a position crossing the route of the host vehicle;
in a case in which the aspect of the accident corresponds to a left-turn accident at the intersection, display the object simulating a pedestrian at a position crossing the route of the host vehicle; and
in a case in which the aspect of the accident corresponds to a right-turn accident at the intersection, display the

object simulating another vehicle traveling in the opposite direction at a position of an oncoming lane of the host vehicle.

* * * * *